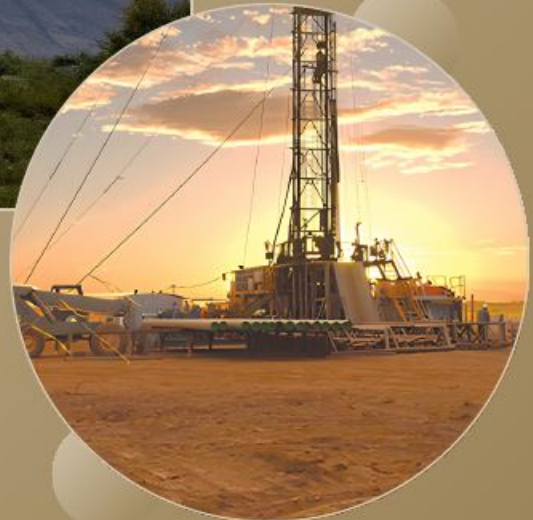


Sweetwater Helium llc Proposal



Prepared for

Navajo Nation

by

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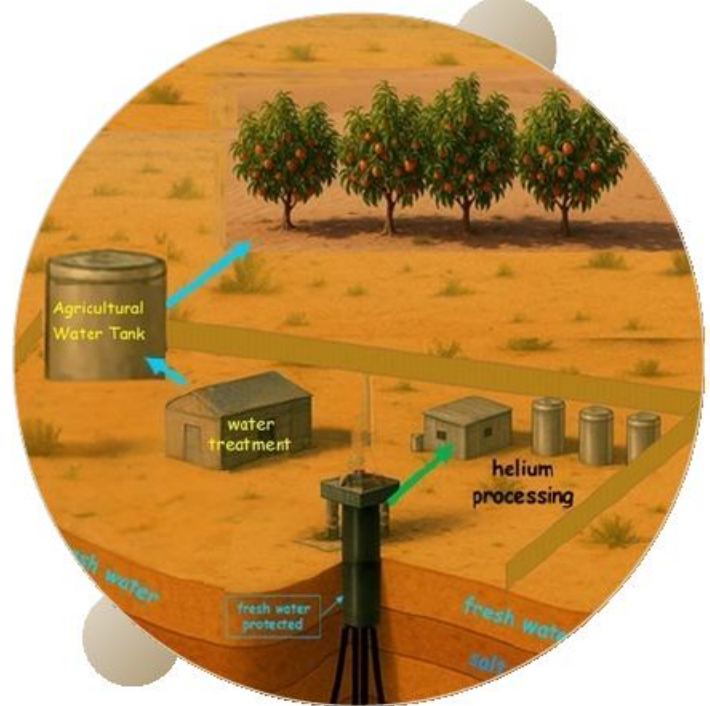


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Executive Summary

Sweetwater Helium llc

Helium & Agricultural Water Project



Traditional Navajo stories and geologists agree the Navajo land was once covered by vast oceans. That ocean water is still trapped in the rock thousands of feet below the surface. Millions of years ago, nitrogen and helium from hundreds of miles underground were brought to the surface by volcanic eruptions. This is the same nitrogen that makes up 80% of the air we breathe. As it moved upward, these gases mixed with the ancient seawater already trapped in the rocks. Because helium and nitrogen don't dissolve well in water, they form tiny bubbles, much like the fizz in a carbonated drink. We will drill and produce both the mixed gases and the salty water. We will separate the helium and nitrogen from the water and re-inject most of it back into the formations from where it came. Ultimately, some of this water will be desalinated into fresh agricultural water.

Key proposal highlights include:



Produce ancient seawater and ancient air from deep wells east of Black Mesa.



Recover helium from the produced ancient air for commercial sale.



Desalinate produced seawater and deliver fresh water for agricultural use.



Provide a share of revenue from helium production to the participating Chapters.

The project is designed to create many benefits:
a new helium resource, revenue, jobs, and a new agricultural water source for the Navajo Nation

Project Overview



This is a conceptual illustration, not a final engineering design.





This is a conceptual illustration, not a final engineering design.

What We Know So Far

Past oil exploration at the proposed sites did not identify commercial oil or hydrocarbon gas, but the wells provided valuable geologic information. They confirmed deep, ancient saltwater containing naturally occurring gases more than a mile below the surface. We know the gas is 85 to 90% nitrogen with a little CO₂, but we believe it may also contain helium. The helium (and nitrogen) being evaluated is believed to be primordial — ancient helium trapped during Earth's formation — rather than helium created by radioactive decay. Neither nitrogen or helium are not very soluble in water so, they will naturally separate into tiny bubbles — like opening a carbonated beverage. Helium can then be collected from this released gas stream. The helium being evaluated is believed to be primordial helium — ancient helium trapped during Earth's formation — rather than helium created by radioactive decay.

Sweetwater's work is to determine whether that gas contains enough helium to support responsible recovery, while returning most produced water to deep, sealed formations and potentially treating a smaller side stream for agricultural use

Helium is a critical element used in medical imaging, semiconductor manufacturing, aerospace, fiber optics, and scientific research. The demand for helium is growing and has the potential to become a significant new source of revenue for the Nation

Some Facts



Old wells confirm no commercial oil and gas in the proposed area.



Helium and oil have different origins and do not have to occur together.

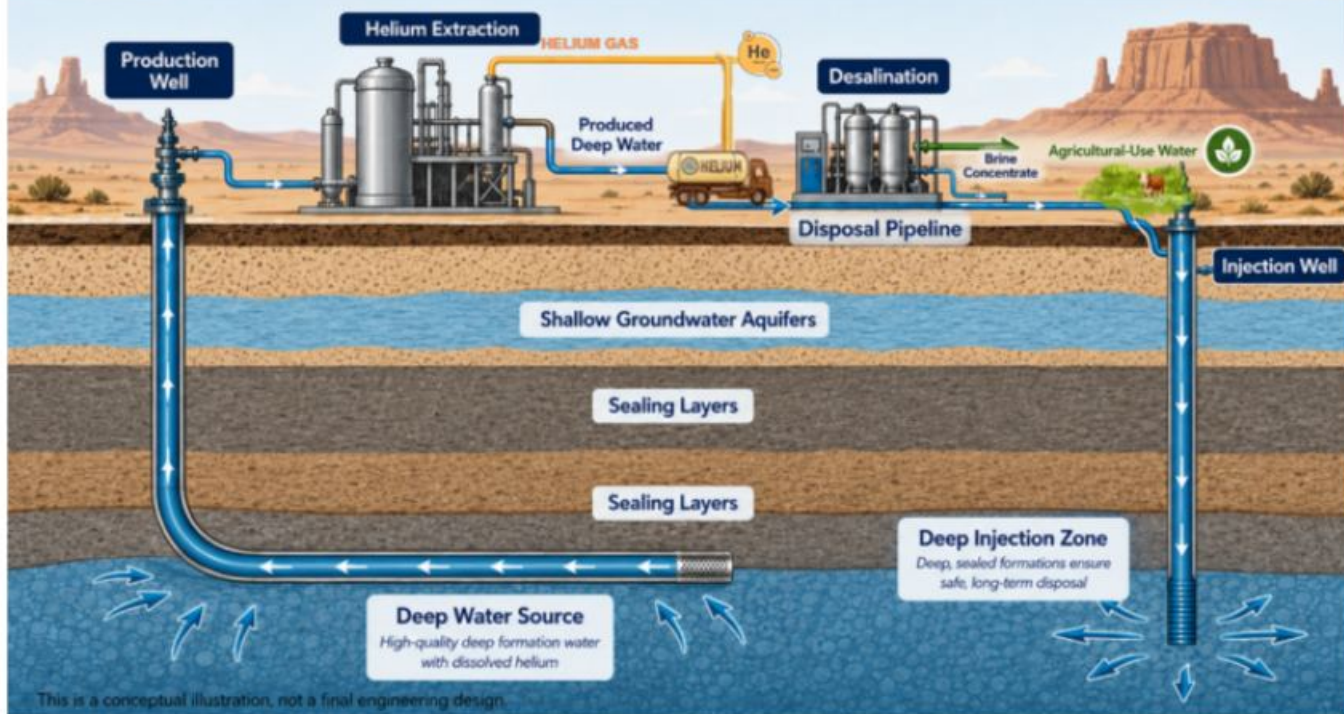


Produced water can be desalinated and reused for agriculture.

This proposal is intended for Chilchinbeto, Dennehotso, and Mexican Chapters of the Navajo Nation.

Sweetwater Helium: Deep Water Use Concept

Most deep water is reinjected. A smaller treated side stream may support agricultural-use water.



Proposal & Technical Scope

01

Recover helium from ancient air trapped in deep underground

02

Remove salt from produced seawater for agricultural use.

03

Return majority of water. Deliver agricultural water for irrigation.

04

Share revenue from helium production with the Nation

The Process

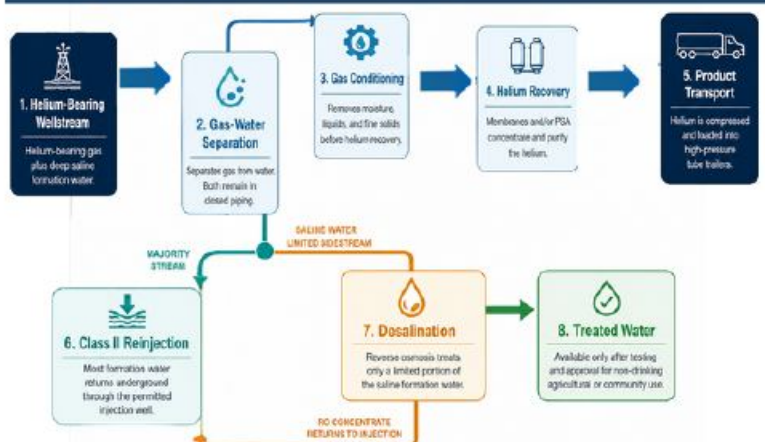
Helium is usually recovered from underground gas reserves that contains helium mixed with nitrogen and other gases. When the gas reaches the surface, equipment first removes water, all moisture, and solid particles. The gas may then pass through membrane units that separate gases according to how quickly they move through the material. Helium passes through the membrane more readily than many of the other gases, creating a more concentrated helium stream. Additional purification may be performed using

pressure-swing adsorption, or PSA. In a PSA system, the gas is placed under pressure and passed through special materials that capture impurities such as nitrogen, carbon dioxide, and moisture while allowing the helium-rich gas to continue through the system. The pressure is then lowered so the captured gases can be released and the material can be reused.

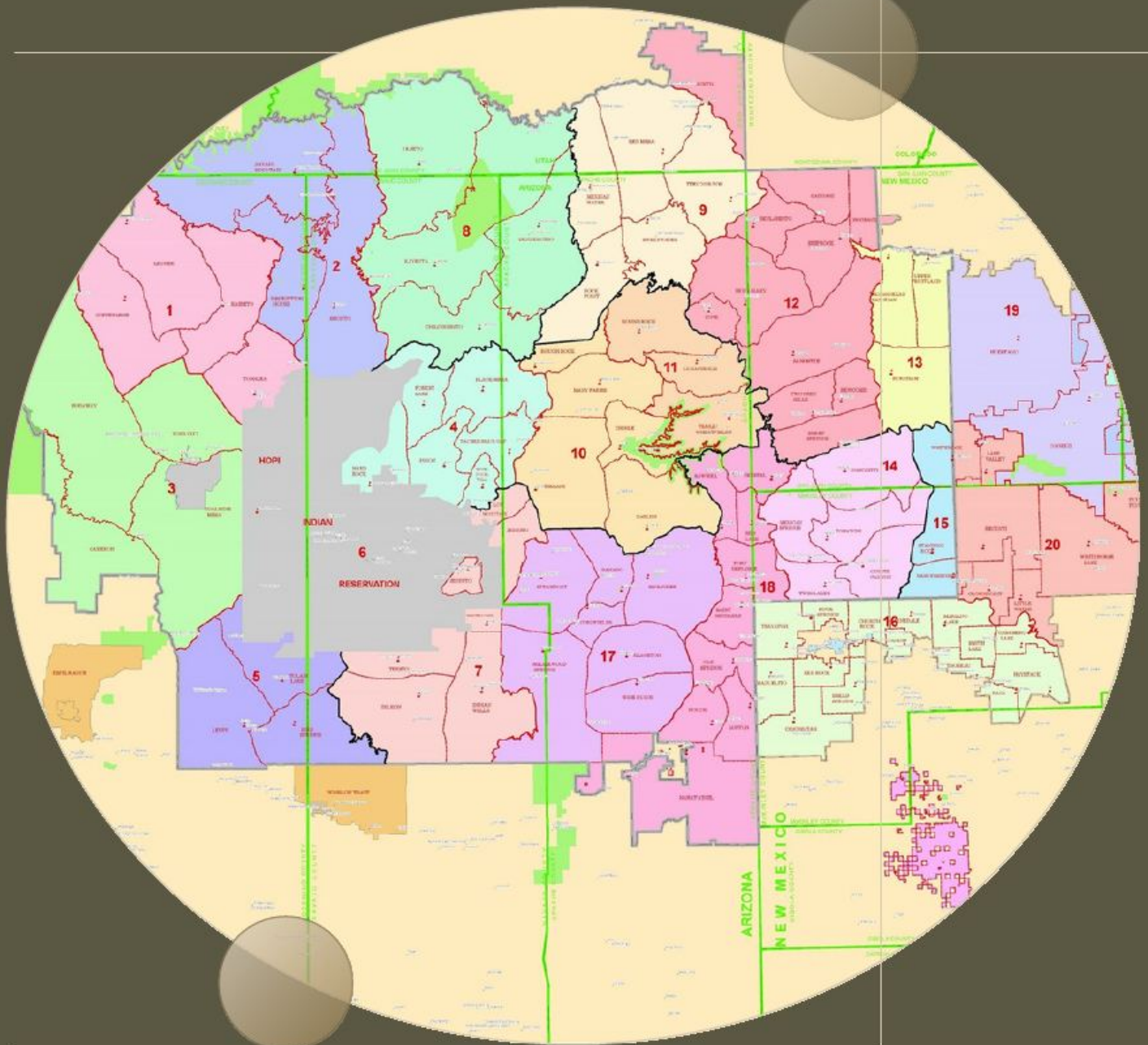
Membranes and PSA may be used separately or together, depending on the composition of the gas and the purity required. Cryogenic equipment, which uses extremely low temperatures, may be added to producing very high-purity Grade-A helium or preparing helium for liquefaction.

SWEETWATER'S CLOSED GAS-AND-WATER HANDLING SYSTEM

THE WELLSTREAM IS SEPARATED AT THE SURFACE: HELIUM IS RECOVERED AND TRANSPORTED, WHILE SALINE FORMATION WATER STAYS IN A CLOSED TREATMENT AND REINJECTION LOOP.



Proposed Area of Interest



Proposed site Locations

Sweetwater Helium

Chilchinbeto • Dennehotso • Mexican Water



1 Chilchinbeto

PROPOSED SITE

Latitude 36.678752° N
Longitude 110.084389° W

2 Dennehotso

PROPOSED SITE

Latitude 36.809021° N
Longitude 109.824190° W

3 Mexican Water

PROPOSED SITE

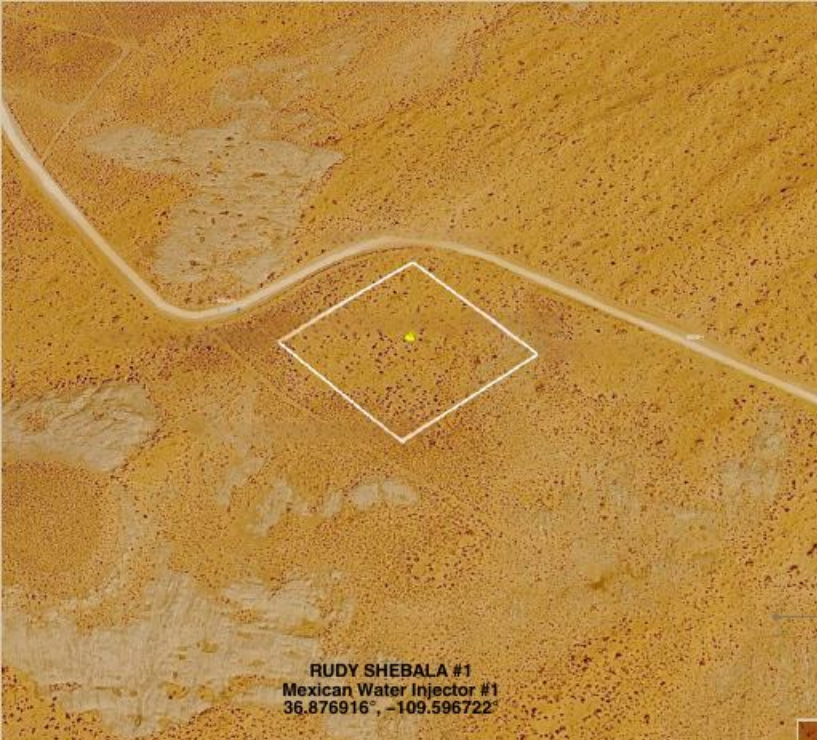
Latitude 36.876916° N
Longitude 109.596722° W

These proposed sites locations are near the Chilchinbeto, Dennehotso, and Mexican Water Chapters. The above map shows each location's relationship to nearby communities and existing roads. The sites provide access to the target formations while keeping the surface footprint as compact as practical. Existing road access reduce the need for new construction and additional land disturbance. Site planning emphasizes groundwater protection, careful water handling, reduced traffic, respect for cultural and biological resources, and ongoing coordination with the nation.



01 Mexican Water

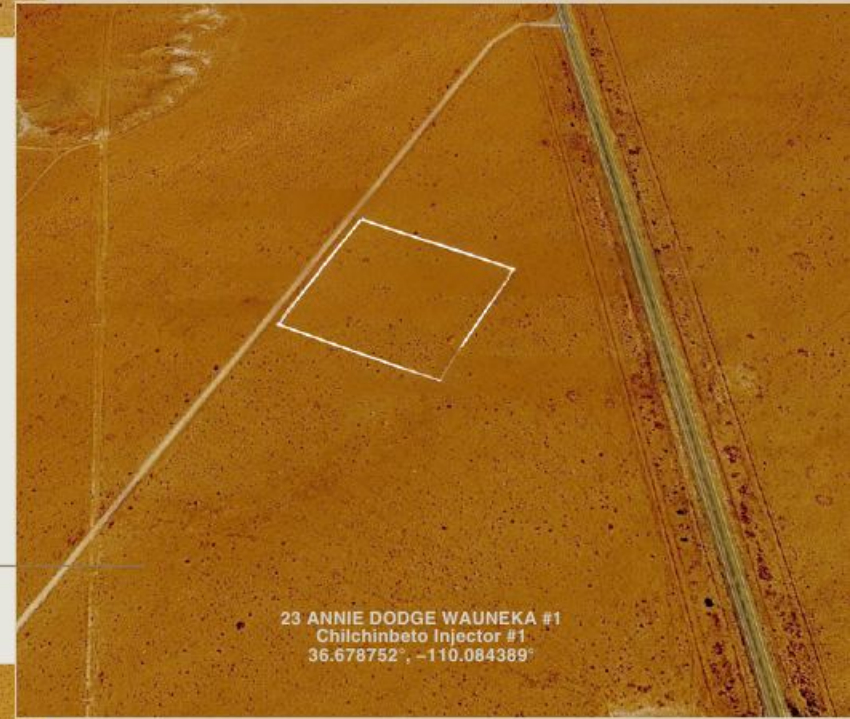
Near existing road access, this proposed approximately three-acre site is planned to limit new disturbance. Final well, pad, utility, and equipment locations remain preliminary and subject to survey, engineering, and Navajo Nation review.



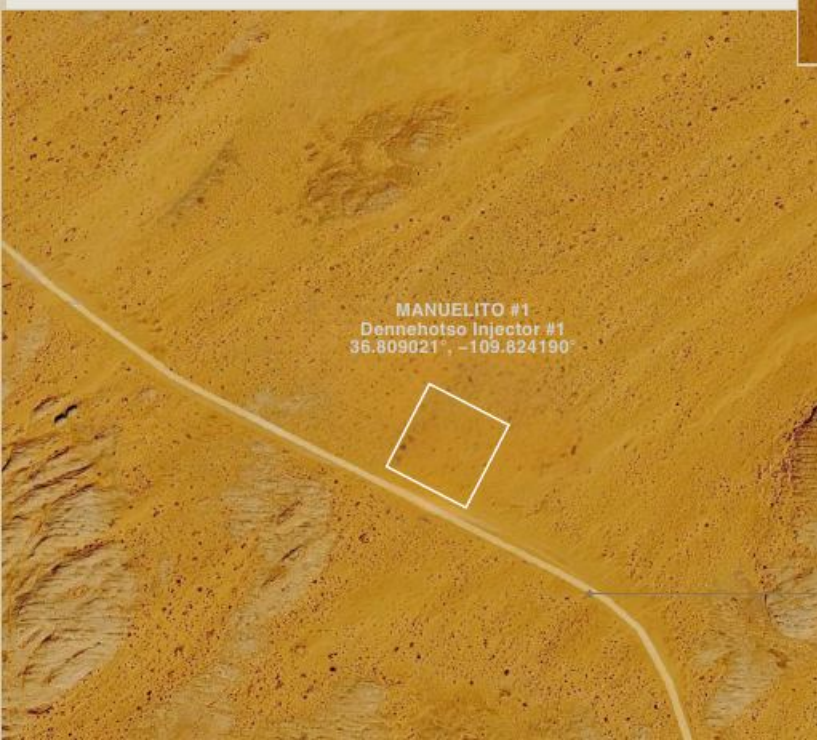
RUDY SHEBALA #1
Mexican Water Injector #1
36.876916°, -109.596722°

02 Chilchinbeto

This proposed approximately three-acre site would support closed gas-and-water handling within a small compact footprint. Existing access would be used where practical. Final placement remains subject to land, biological, cultural, environmental, and engineering review.



23 ANNIE DODGE WAUNKA #1
Chilchinbeto Injector #1
36.678752°, -110.084389°

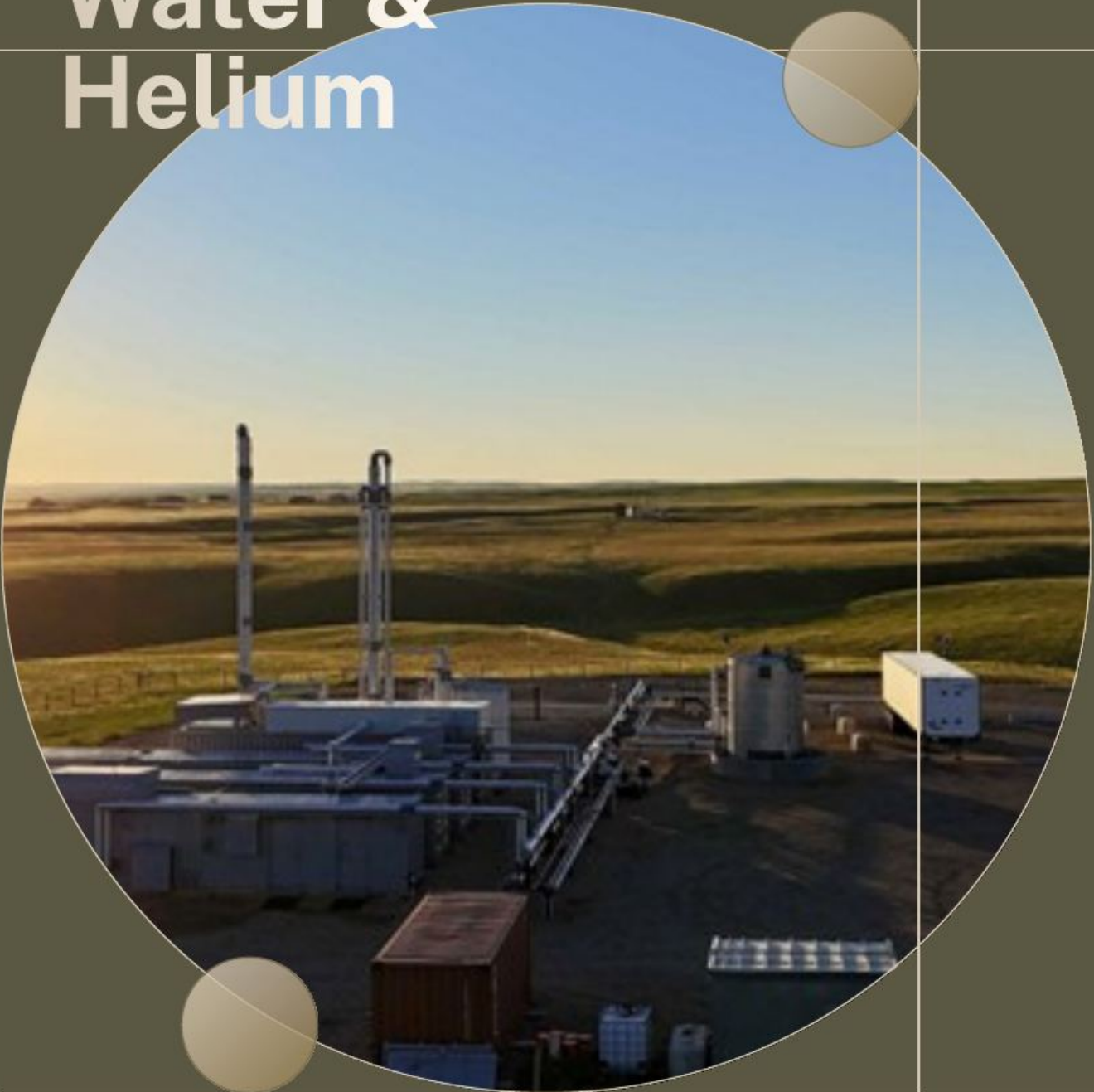


MANUELITO #1
Dennehotso Injector #1
36.809021°, -109.624190°

03 Dennehotso

Near existing road access, this proposed approximately three-acre site is planned to maintain a compact footprint and limit new surface disturbance. Final well, pad, utility, and equipment locations remain subject to survey, engineering, and Navajo Nation review.

Ancient Ocean Water & Helium



This is a conceptual illustration, not a final engineering design.

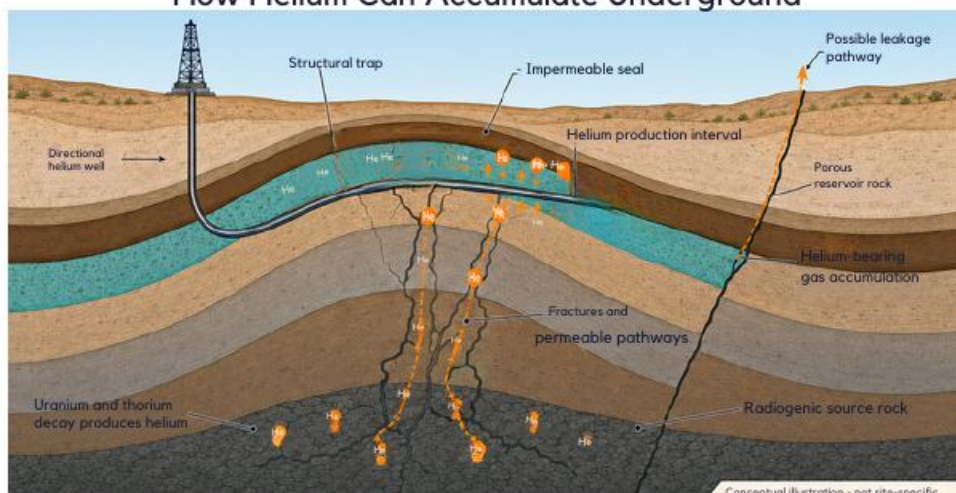


The Source of Helium

Primordial helium is ancient helium trapped within the Earth during its formation, rather than helium created by radioactive decay. Over billions of years, it can slowly migrate upward and accumulate in geologic traps, even where oil and natural gas are absent.

Oil and natural gas form from decomposed organic matter, while helium has a completely different origin. For that reason, helium and hydrocarbons are not necessarily found together. Legacy wells at the proposed sites found no commercial hydrocarbons but identified deep formations containing naturally occurring air, supporting further evaluation for helium.

How Helium Can Accumulate Underground



Source Rocks

Source rocks with high uranium and thorium concentrations generate helium.



Release

Heating and fluid interaction release trapped helium from minerals into fluid and gas phases.

>> Migration Pathways

Faults, fractures, and permeable rock allow helium to travel from the source to a reservoir.



Seals and Traps

Tight seals—such as salt, clay, and shale—can trap helium or slow its movement.



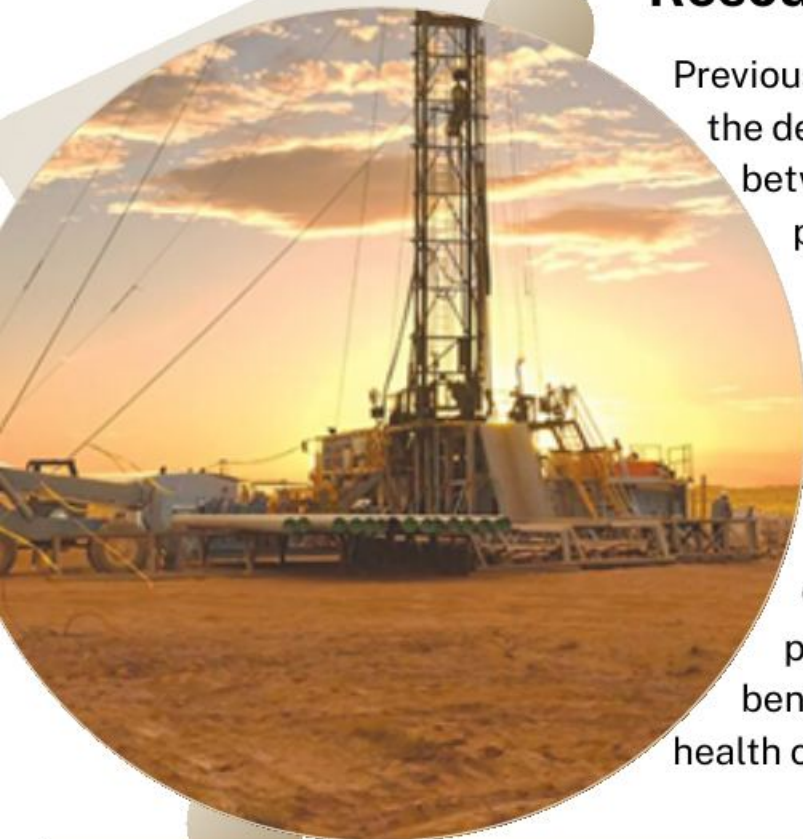
Time Matters

Helium forms slowly. Older source rocks and long-lived, well-sealed traps have had more time to generate and accumulate helium.

The Five Conditions for a Good Helium Reservoir

Resource Insights

Previous wells provided valuable information about the deep geology, saline formations, and separation between shallow groundwater and the proposed production zones. SweetWater has used this information to design protective casing and cement systems, minimize surface disturbance, and plan safe water handling. New drilling and testing would confirm helium concentrations, water quality, and production potential. Clear reporting to the Chapters would remain central throughout the project. The goal is to provide meaningful local benefits while protecting the land, water, and health of nearby communities.



What Helium Is Used For



MRI Systems

Helium is used in medical imaging systems that require very low temperatures.



Scientific Research

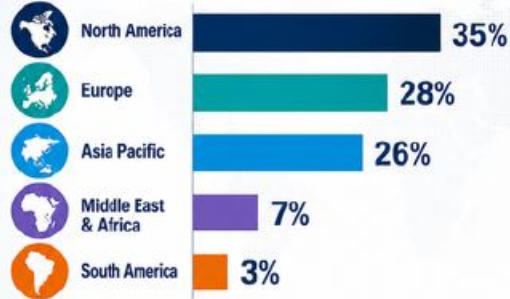
Research laboratories use helium for cooling, controlled environments, and specialized experiments.



WHERE HELIUM IS USED GLOBALLY

Regional Helium Use / Market Demand – NOT Production

Approximate share of global helium use by region



U.S. domestic helium consumption:

56.0
million m³
in 2023

WHAT HELIUM IS USED FOR



HEALTHCARE / MRI

Essential for MRI cooling and other medical applications.



SEMICONDUCTORS & ELECTRONICS

Critical for chip manufacturing and advanced electronics.



AEROSPACE & DEFENSE

Used in rocket pressurization, testing, and space exploration.



RESEARCH & CRYOGENICS

Enables low-temperature research and scientific breakthroughs.



LEAK DETECTION & ADVANCED MANUFACTURING

Powers helium leak detection and specialty industrial processes.



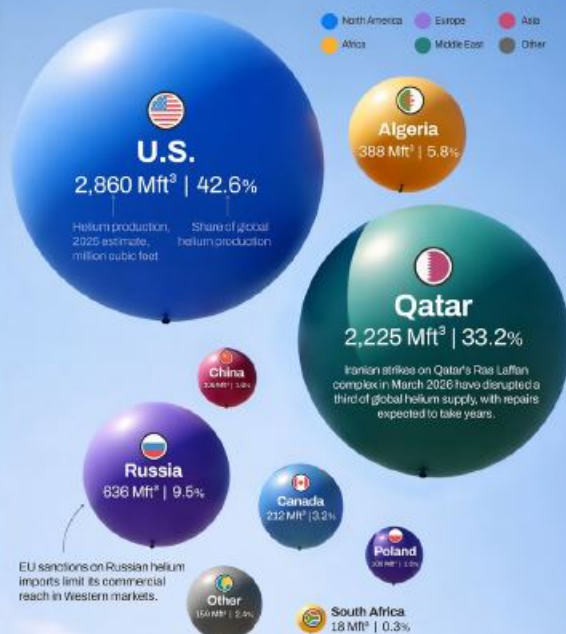
IMPORTANT

Countries such as Qatar and Algeria are major helium producers and exporters, but the largest USE is concentrated in regions with major healthcare, semiconductor, research, and advanced manufacturing industries.

Helium Production

by Country in 2025

North America Europe Asia
Africa Middle East Other

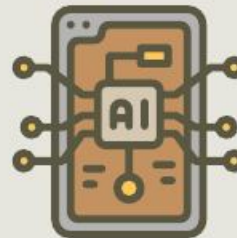


Share of Global Helium Consumption



U.S. Helium Reserve is an estimated 1.2 billion cubic feet. Helium is a non-renewable resource. SOURCE: U.S. Geological Survey Mineral Commodity Summaries 2025

VISUAL CAPITALIST



High-Tech Manufacturing

Helium supports fiber optic manufacturing, computer chip production, welding, and leak detection.



Aerospace & Safety Uses

Helium is used in breathing mixtures, high-energy lasers, and lifting heavy payloads.



Weather Forecasting

Helium lifts weather balloons that collect atmospheric data for weather forecasting, storm tracking, and climate research.

Helium Processing



Helium Processing Proven through Practice

Helium is typically extracted from a gas mix. Initially, the well stream undergoes separation and dehydration processes to eliminate produced water, moisture, and solids. Following this, membrane systems and/or pressure-swing adsorption (PSA) are employed to concentrate the helium.

This separation is a physical process, not a chemical reaction. The water is not chemically altered to produce helium; the existing helium and nitrogen simply come out of solution as pressure decreases.

Helium plants generally utilize a combination of PSA and membrane technologies, while facilities producing crude and Grade-A helium rely on cryogenic extraction.

The key steps include:

- Extracting gas from produced water.
- Removing moisture and impurities.
- Concentrating helium via membranes and/or pressure-swing adsorption.
- Purifying and compressing the gas for transportation.
- Use cryogenic methods only when higher purity or liquid helium is needed.

Historically, commercial helium has been recovered from fields in Kansas, Oklahoma, Texas, Colorado, Utah, and Wyoming. Today, modern facilities also operate in Canada and other regions around the world

Sources: BLM Helium Fact Facts; USGS Minerals Yearbook.

This is a conceptual illustration, not a final engineering design.



Because the facility will use electric power, normal operations are expected to produce low noise levels. The primary sound sources will be enclosed compressors, pumps, cooling fans, and periodically cycling process valves. Noise-control measures — including equipment enclosures, silencers, vibration isolation, and appropriate building insulation — will be incorporated into the final design.

Helium Processing Pathway

Helium-bearing gas is separated, conditioned, concentrated, purified, and compressed; produced water is handled separately.

HELIUM PURIFICATION MODULE

Representative equipment - final configuration subject to detailed engineering



1 FEED-GAS INLET & ISOLATION – Receives the helium-bearing gas stream and allows safe shutoff.

2 INLET SEPARATOR – Removes entrained liquid droplets and condensate before compression.

3 PRETREATMENT FILTERS – Remove particles, oil aerosols and residual moisture.

4 GAS COMPRESSOR – Raises gas pressure for efficient separation and purification.

5 PURIFICATION BANK – Membrane or PSA vessels separate helium from nitrogen and other gases.

6 HEAT EXCHANGER / COOLER – Controls gas temperature between process stages.

7 HELIUM BUFFER VESSEL – Stabilizes purified-helium pressure and product flow.

8 PLC CONTROL & EMERGENCY STOP – Monitors the process and provides automatic safety shutdown.

9 PIPING, VALVES & INSTRUMENTS – Route flow and monitor pressure, temperature and flow rate.

Sweetwater's proposed system would use modular, enclosed equipment selected for the actual gas composition and production rate. Initial on-site processing would focus on producing transportable gaseous helium. Liquefaction, if ever used, would be a separate later-stage decision. Process controls, gas detection, pressure protection, leak monitoring, and emergency shutdowns would be incorporated into the final facility design.

Desalination



Desalination

A Practical, Established Technology

Desalination is the process of removing dissolved salts and minerals from seawater, brackish groundwater, or other saline water to produce usable fresh water. It is now a well-established technology used for municipal drinking water, agriculture, industry, mining, and emergency water supplies.

Desalination is increasingly common worldwide. More than 20,000 desalination plants operate in over 150 countries, supplying fresh water to approximately 300 million people every day. Global desalination capacity has grown by an average of roughly 7% per year since 2010.



Desalination is especially important in places with limited rainfall or dependable freshwater supplies. Major users include:

- Saudi Arabia and the Persian Gulf states, which produce nearly half of the world's desalinated water.
- Singapore, which currently operates five desalination plants.
- Australia, where desalinated seawater commonly provides 30–40% of Perth's drinking water.
- Spain, China, India, Chile, California, Florida and Texas, where desalination supports cities, industries and agriculture.
- Texas, which has 60 municipal desalination facilities, including 43 that treat brackish groundwater.

Sources: USGS; Texas Water Development Board (2020); PUB Singapore.

Desalination Water Pathway

Reverse Osmosis separates deep saline produced water into two controlled streams: treated water for agricultural use, and concentrated brine for deep Class II reinjection. Desalination is expected to continue expanding as drought, population growth, groundwater depletion and climate change place additional pressure on conventional water supplies. Improving membranes, energy-recovery equipment and renewable-energy systems are reducing operating costs. Its principal challenges remain energy consumption. Overall, desalination has moved from a specialized technology used mainly in extremely dry countries to an increasingly important part of the world's regular water-supply infrastructure.



- 1 **FEED-WATER INLET & ISOLATION** – Receives saline water and allows safe shutoff of the treatment module.
- 2 **PRETREATMENT FILTER BANK** – Removes suspended solids and protects downstream equipment.
- 3 **CHEMICAL DOSING SYSTEM** – Adds antiscalant or adjusts pH when required for membrane protection.

- 4 **CARTRIDGE SAFETY FILTER** – Captures fine particles immediately before the high-pressure pump.
- 5 **HIGH-PRESSURE PUMP** – Raises feed-water pressure for reverse-osmosis separation.
- 6 **RO MEMBRANE VESSELS** – Separate dissolved salts from the water under pressure.

- 7 **PRODUCT-WATER OUTLET** – Delivers treated non-potable water for approved beneficial use.
- 8 **CONCENTRATE / BRINE OUTLET** – Routes the concentrated salt stream for controlled reinjection.
- 9 **PLC CONTROL & EMERGENCY STOP** – Monitors pressure, flow and water quality and provides automatic shutdown.

REVERSE-OSMOSIS DESALINATION MODULE
Representative equipment - final configuration subject to detailed engineering

Well Design & Groundwater Protection



Groundwater Protection

Groundwater Protection Beyond Standard Practices

Our well design prioritizes groundwater protection and is engineered to exceed conventional well-construction standards. It features multiple steel casing strings. This will include approximately 3,000 to 3,600' of surface casing set below the base of the freshwater table. An additional protective casing extending from the bottom of the well to the surface (approximately 6,200 to 7,200') will add additional barriers between shallow aquifers and the deep production zone. This advanced, six-layered system offers superior protection. By:

- Isolating geological formations
- Containing produced fluids
- Reducing potential pathways to groundwater

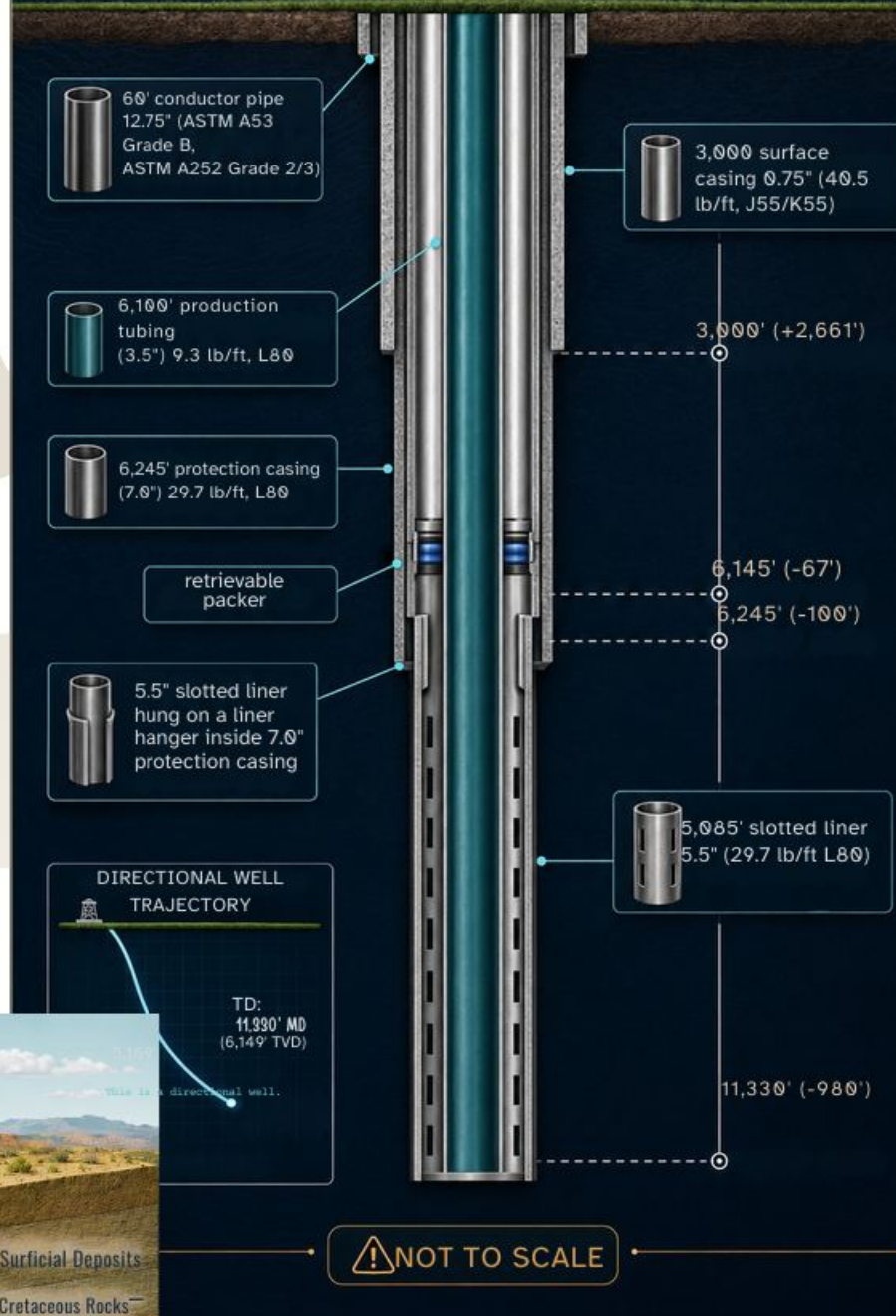
To ensure the system's long-term integrity and environmental performance, we will conduct cement verification, pressure testing, continuous monitoring, and regulatory reviews.

Sample Surface Casing

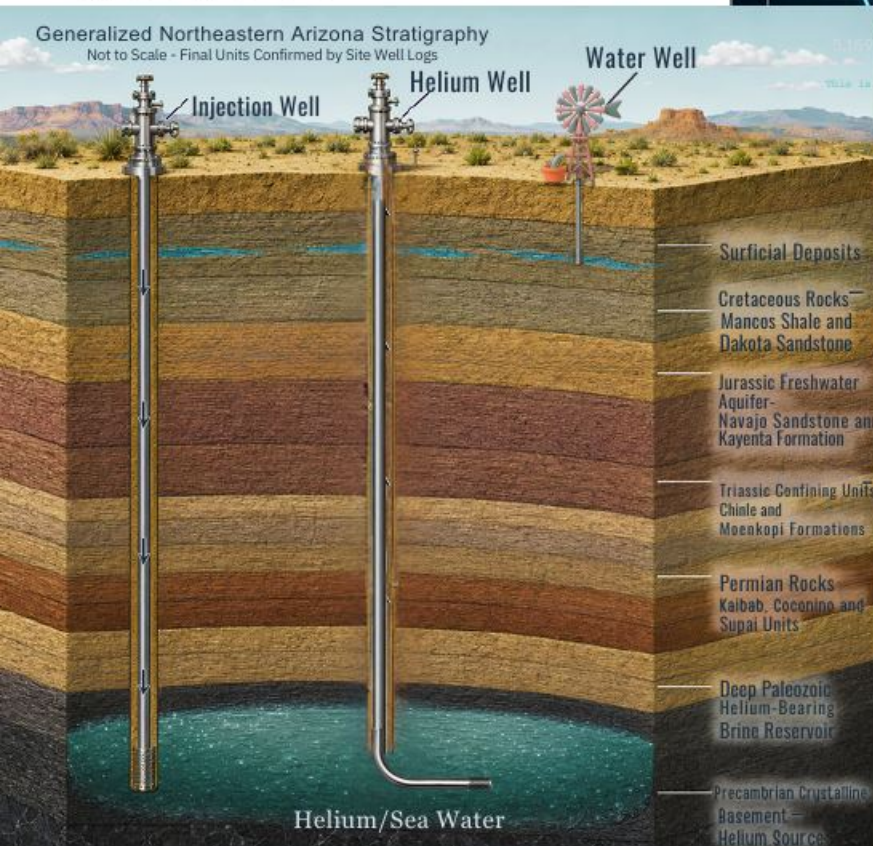
Elevation: 5,169' Total Depth: 11,330' MD (6,149' TVD)

Dennehotso Chapter

Note: This is a directional well



Generalized Northeastern Arizona Stratigraphy
Not to Scale - Final Units Confirmed by Site Well Logs



NO FRACKING

PROTECT OUR LAND. PROTECT OUR WATER. PROTECT OUR FUTURE.

Our Position on Fracking

Sweetwater Helium **does not use fracking**. We are **against fracking**, have **never used it**, and **never will**.

Our project is designed around **responsible helium recovery**, **water management**, and **protection of land and groundwater**.

The Proposed Facility & Location



Facility

Sweetwater's proposed facility integrates helium recovery, water treatment, injection, and transportation on one compact, secure pad near the production wells. Processing equipment will be housed inside the operations building to reduce weather exposure, noise, and unauthorized access. Closed piping will transfer gas and water between the wells, treatment systems, storage, and loading areas. The layout supports continuous monitoring, efficient truck circulation, and rapid emergency shutdown while minimizing the surface footprint.

Facility Plan

CONCEPTUAL SWEETWATER FACILITY

A Well Production

Helium-bearing gas and saline formation water are brought to the surface and routed through a closed gathering system for controlled separation and processing.

B Helium Processing

Membrane separation and PSA concentrate and purity of the helium. Finished product is loaded into high-pressure tube trailers for transport to market.

C Water Treatment

Produced water is degassed, filtered, and tested. A portion may be treated by reverse osmosis for approved non-potable agricultural or community use.

D Initial Separation

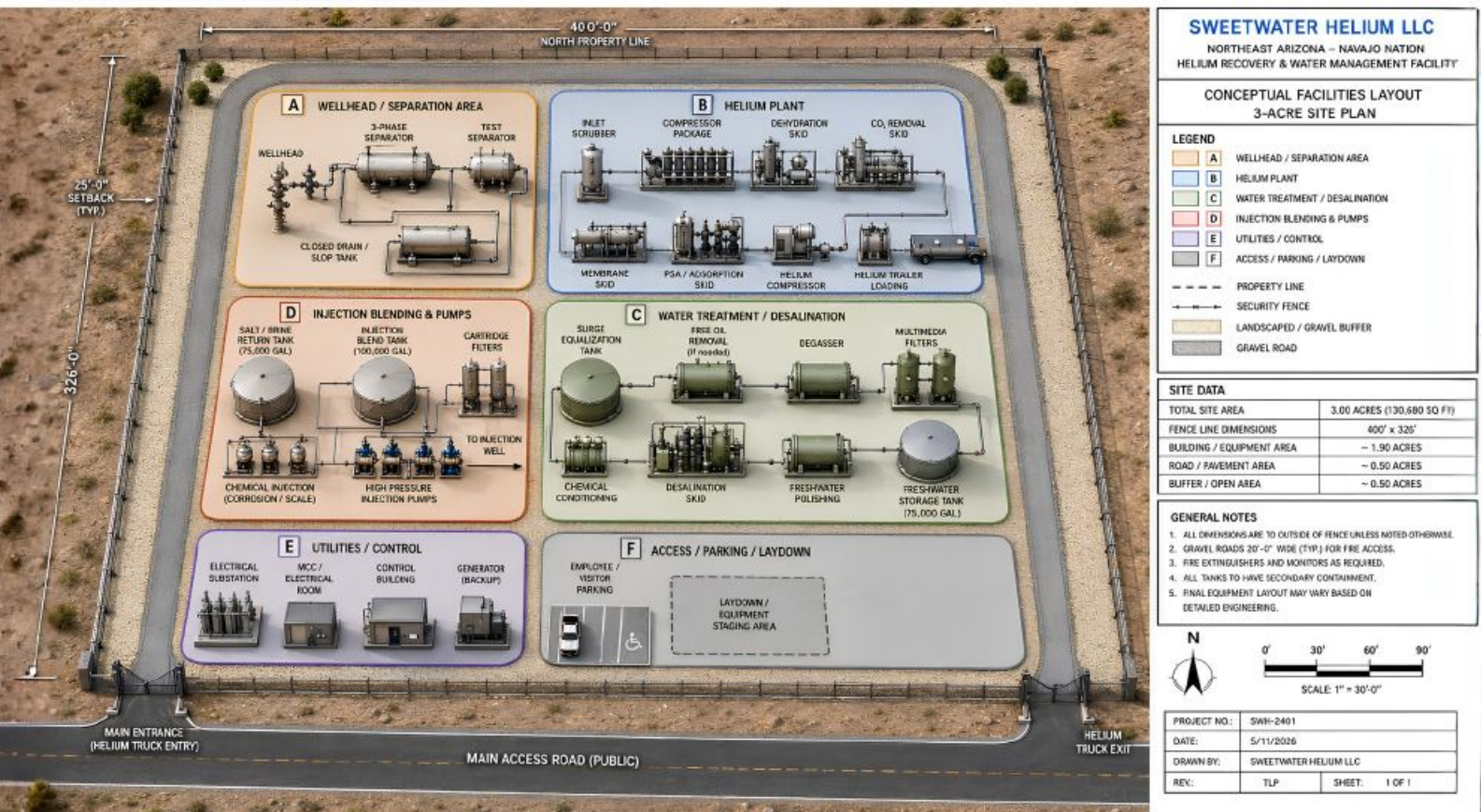
Separators remove produced water, entrained solids, and condensate. The gas is then hydrated and conditioned before entering the helium-recovery process.

E Injection

Untreated produced water and RO concentrate are routed through closed piping to a permitted Class II injection well. No open evaporation ponds are proposed.

F Operations & Oversight

Personnel monitor pressure, flow, water quality, equipment performance, truck loading, and site security. Automated alarms and shutdowns provide additional protection.



The conceptual three-acre plan separates wellheads, processing, water treatment, injection, utilities, loading, and vehicle circulation into defined operating zones. The plan is a functional cutaway; processing systems are enclosed inside the operations building. Final placement remains subject to detailed engineering.

Community Questions. Straight Answers.

What Sweetwater is promising — and what you can expect.



Protecting groundwater

Three cemented steel casings, six layers total, separate the well from groundwater. We pressure-test the well and continuously watch for leaks.



Land disturbance

Each pad is limited to about three acres and uses existing roads.



Community benefits

The stated package includes a \$1 million infrastructure grant per Chapter after the first well, a 5% Chapter net-profit interest, \$60,000 in yearly scholarships, and 6–10 local jobs with training at each operating site.



Air and odors

Helium stays in a closed system. We inspect for leaks and do not vent toxic gas. Electric equipment is used when practical to reduce exhaust and noise.



Handling produced water

Water stays in closed tanks and pipes. Most goes back underground through the permitted injection well. We accept no outside waste and use no open evaporation pond.



Conserving helium

Helium is captured in a closed system and sold rather than intentionally vented or wasted.



Culture and Navajo authority

Cultural, archaeological, biological, grazing, land, and environmental reviews must be complete before work begins. A cultural discovery stops work.



Pressure, old wells, and earthquakes

Before injection, nearby wells are reviewed and a safe pressure limit is set. If pressure or monitoring looks unusual, injection stops.



Plugging and cleanup

Sweetwater pays to plug the wells and restore the sites. Closure money is set aside before drilling.



Water use and treated water

The water comes from deep salty formations — not shallow drinking-water supplies. Most is reinjected. A limited amount may be treated for approved non-drinking agricultural or community use.



What happens if production ends?

Sweetwater remains responsible for plugging, cleanup, and land restoration. The closure fund does not depend on future production.



Noise, lights, dust, and traffic

We use electric equipment, shield lights downward, control dust, manage truck traffic, and repair project-related road damage.



Spills or equipment problems

Containment, alarms, inspections, spill kits, and emergency shutoffs limit releases. Damaged equipment must be repaired and tested before restarting.



No Fracking Ever!

We do not use fracking. Our helium is produced through conventional drilling without hydraulically fracturing the rock.

OUR PROMISE



No fracking or outside waste.



Closed gas and water handling.



No open evaporation pond or toxic venting.



24/7 staffing leak detection monitored



Sweetwater-funded plugging and land restoration.



Sweetwater Helium LLC •
P.O. Box 130771, Spring,
TX 77393



Proposal to the Navajo Nation

Sweetwater Helium LLC

Contact: Tom Pellegrini, President
(928) 307-7824

tom@sweetwaterhelium.com |

tom@pellegrini.us

10800 Gosling Road, Box 130771,
Spring, Texas 77381



INTRODUCING SWEETWATER HELIUM

Yá'át'ééh

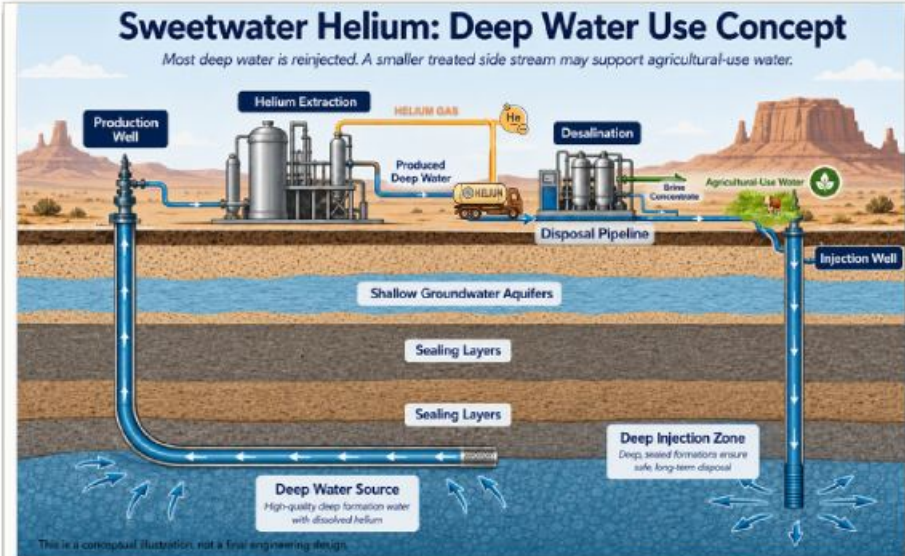
Sweetwater Helium LLC respectfully introduces a proposed helium production and water-management project in the Dennehotso, Chilchinbeto, and Mexican Water areas. Our concept includes a small desalination facility in each area to treat a portion of naturally occurring saline water for community-approved, non-potable uses such as agriculture, livestock watering, and dust control."



We are committed to protecting the land and water while creating meaningful benefits through Chapter-selected infrastructure, local jobs, scholarships, and long-term community participation.

PROJECT SCOPE

- Three proposed helium production sites
- Helium separation for domestic use
- Closed-system produced-water handling
- Desalination for non-potable water
- Reinjection as the primary water-management method



WHAT'S UNDERGROUND AND HOW THE PROCESS WORKS

The helium and nitrogen we produce originated deep underground and were brought to the surface by volcanic eruptions millions of years ago. As they rose, these gases mixed with the ancient seawater already trapped in the rocks. Because helium and nitrogen don't dissolve well in water, they form tiny bubbles, much like the fizz in a carbonated drink.

The salty water is pumped to the surface in a sealed system, where helium and nitrogen are separated. The water is then returned underground, protected by six layers of steel casing and cement to prevent contamination of groundwater.

PROTECTING LAND & WATER



Groundwater protection through strong well design and standards.



Produced-water reinjection as the primary management method.



No open evaporation ponds. Minimized surface, noise and lighting.



Closed-system processing with no surface discharge.



Our project is responsible for helium recovery, water management, and groundwater recovery.

PROPOSED SITE AREAS DENNEHOTSO, CHILCHINBETO, AND MEXICAN WATER



A Partnership Built on Respect

Responsible helium development should protect the land, conserve water, and add value in the communities where it operates.

COMMUNITY BENEFITS



Proposed share for the Nation and each chapter's location



Up to \$1 million per Chapter for selected water infrastructure.



Annual scholarship support totaling \$60,000.



Road maintenance related to project operations.



Local jobs, training and Navajo contractor opportunities.

Community Concerns



No Fracking. Limited Surface Footprint

Sweetwater Helium does not use fracking. We are against fracking, have never used it, and never will. Our project is designed around responsible helium recovery, water management, and protection of land and groundwater.

Spills, poorly maintained sites, & damaged roads

Fluids will be kept in closed tanks and piping with secondary containment to prevent spills from ever reaching the land. Alarms, routine inspections, spill-response equipment, and emergency shutdown systems will be in place. The site will be staffed 24/7, and Sweetwater will promptly clean up any incident and repair roads damaged by project operations.



Groundwater Contamination

Only water produced by the project would be reinjected. Multiple layers of steel casing and cement, along with sealed tubing and packers, would keep the water isolated from usable groundwater. The wells would undergo pressure and integrity testing and be continuously monitored for leaks. No open evaporation ponds would be used.

Abandoned wells and unfunded cleanup

Each site will have a detailed plugging-and-closure plan before drilling begins. Sweetwater will be responsible for all closure costs, with funds placed in a non-revocable trust in advance so the wells can be properly plugged and the sites restored.



Residents, grazing permit holders, and cultural places excluded from decisions

Sweetwater will provide regular updates to Chapters, grazing permit holders, nearby residents, and other affected community members. We will share major milestones, upcoming fieldwork, and project changes, with clear ways to ask questions and raise concerns.



Air pollution, open casing vents, odors, and delayed leak repair

Gas will be handled in a closed system with no toxic venting. LDAR inspections, optical gas-imaging cameras, and odor tracking will help identify leaks early. Trained staff will be on site 24/7 and authorized to stop operations immediately if a problem is detected.



Key Proposal Highlights

HELIUM RECOVERY + WATER BENEFITS INITIATIVE



Protecting the Land. Supporting the Community.

Sweetwater Helium believes responsible development must protect the land, conserve water, and create lasting benefits for the communities where we operate.

Our approach is designed to minimize surface disturbance, prevent unnecessary emissions, safeguard groundwater, and responsibly manage produced water. We are committed to working openly with Navajo Nation leaders, local Chapters, grazing permit holders, and community members throughout every stage of the project.

Success will not be measured only by the helium we produce, but by the jobs we create, the infrastructure we support, the water resources we help develop, and the trust we earn.

We welcome your questions, guidance and candid feedback.



Strategically located in the Southwest to power critical industries and strengthen America's supply chain.



P planned/proposed E expanding O operating

Reduces U.S. reliance on imports and vulnerable foreign supply routes.
Strengthens national security and supply chain resilience.



Helium is essential for chip manufacturing—cooling, cleaning, and leak detection
SEMICONDUCTORS

U.S. semiconductor manufacturing capacity is projected to increase about 203% between 2022 and 2032.



Vital for aerospace, satellite, missile, and defense systems. Supports stealth tech, hypersonics, and high-purity manufacturing operations



weetwater's produced water creates a platform for valuable mineral recovery and responsible water reuse in a water-scarce region.



The core investment is domestic helium production located near major semiconductor, aerospace, and defense markets in an already tight and supply-constrained market. Water reuse (desal), mineral recovery, and helium-3 could provide additional long-term potential, but the project's current value is supported by helium production alone

CRITICAL & IRREPLACEABLE — Helium's unique properties make it irreplaceable in high-tech manufacturing, scientific research, healthcare, and national defense.



DOMESTIC SUPPLY GAP — The U.S. relies heavily on imports. Limited domestic production creates vulnerability to global disruptions and export restrictions.



STRATEGIC ADVANTAGE — Domestic helium strengthens U.S. industrial leadership, supports national security, and secures supply for growing technology industries.

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HELIUM Helium-3 is excluded from current revenue projections, as it represents a long-term opportunity related to fusion energy and space exploration. Its scarcity and potential applications give it a theoretical value of \$800,000 to \$1 million per billion cubic feet (Bcf) of raw gas, but this is not included in current project economics.



MINERAL RECOVERY — Produced water is a resource. Sweetwater is advancing technologies to recover minerals from oil- and gas-produced water while enabling responsible water reuse in the Southwest. *Potential future value — not included in current project economics.*



U.S. Industrial Growth Will Drive the Next Helium Demand Cycle

+203%

SEMICONDUCTORS
U.S. semiconductor manufacturing capacity will increase between 2025 and 2032.

6–10%

HISTORICAL HELIUM
Historical helium concentrations reported across Sweetwater project areas.

35+Bcf

EST. RECOVERABLE
HISTORICAL HELIUM
concentrations reported across Sweetwater project areas.

+36%

Domestic Demand
Expected minimum U.S. helium demand increase by 2035

Price Volatility & Supply Risk

The global helium market may experience near-term rebalancing as new supply enters production. Prices nevertheless remain elevated by historical standards, while concentrated production, geopolitical exposure and a limited number of major suppliers continue to create supply uncertainty.

At the same time, U.S. semiconductor manufacturing, aerospace and other advanced technologies are expanding rapidly. Sweetwater's target 2029 production start aligns with this new industrial capacity coming online.

Sweetwater does not depend on a permanent helium shortage or continued price spikes. The opportunity is to establish a long-life domestic helium source near major Southwest demand growth, providing customers with greater supply security as semiconductor, AI and aerospace requirements expand through the 2030s.

Potential Upside Not Included



The +36% demand scenario is based primarily on projected semiconductor and aerospace growth. It does not assume additional helium demand from emerging technologies such as quantum computing or other advanced cryogenic applications. Quantum systems require extremely low temperatures, which rely on helium

Why Domestic Supply Matters



Helium supply remains concentrated among a small number of global producers. A long-life U.S. source near major Southwest semiconductor, aerospace and AI investment would provide customers with greater supply security and shorter domestic logistics.



2028
Dry Piney, Wyoming expected to add more than 0.8 Bcf/year of new U.S. helium capacity



2028-2030
Major new semiconductor fabs and supporting industrial-gas infrastructure enter operation



2028
Sweetwater target first production



2030s
Semiconductor and aerospace growth will progressively absorb today's new helium capacity